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Subject: DBMS
Course: B. Tech CSE

Ans 1) A database management system is software that allows user to create, store, manage, retrieve and manipulate data in a structured, efficient way.

DBMS is req because

- Reduces data Redundancy
- Avoids data inconsistency
- Better data security
- Backup and Recovery data integrity
- Data sharing and concurrency control

Ans 2) DBMS is better than the traditional file system because

- Reduces data Redundancy
- Provide data security
- Backup & Recovery
- Easy data retrieval

Ans 3) Data abstraction in dbms is the process of hiding complex internal details of how data is stored and maintained and showing only the necessary information to users.

Levels of data abstraction:

① Physical level (Internal level):

- The lowest level of abstraction
- Describes how data is actually stored in memory.
- Includes details like storage structures, indexing, files organization etc.

② Logical level (Conceptual level)

- Describes what data is stored in the database
- Defines relationship among data
- Does not show physical storage details.

③ View level (External level)

- The highest level of abstraction
- Shows only req data to Sp
- Different users can have diff views of the same database.

Ans 4) The three schema Architecture is the database design framework that separates the database system into 3 diffn levels to achieve data abstraction and independence.

The three schema are

- (i) External Schema (view level)
 - describes how individual users view the data
 - Diffn users can have diffn views
 - hides the rest of the database.
- (ii) Conceptual Schema (logical level)
 - Describes the overall structure of the database
 - Includes tables, attributes, relationship & constraints.
 - Does not include physical storage details.
- (iii) Internal Schema (Physical level)
 - Describes how data is physically stored in memory.
 - Including indexing, storage structure, file organization etc.

* why is it important.

- Provides data abstraction
- Ensures data independence.
- Improves security

Ans 5) Data independence is the ability to modify the database schema at one level without affecting the schema at the next higher level.

* Why is it needed?

- Easier database maintenance
- Flexibility
- Reduced development cost
- System stability

Ans 6) The Entity Relationship model is a high level conceptual data model used to design and represent the structure of a database visually. It was introduced by Peter Chen in 1976.

The ER model describes data using:

- Entities
- Attributes
- Relationships

7) Integrity constraints are rules in a dbms that ensure the accuracy, validity and consistency of data stored in a database. They prevent invalid data from being entered into tables.

Why integrity constraints important? -

- Maintain data accuracy
- Ensure consistency
- Improve Reliability
- Prevents data corruption

Section - B

Ans 1) External Schema (view level):

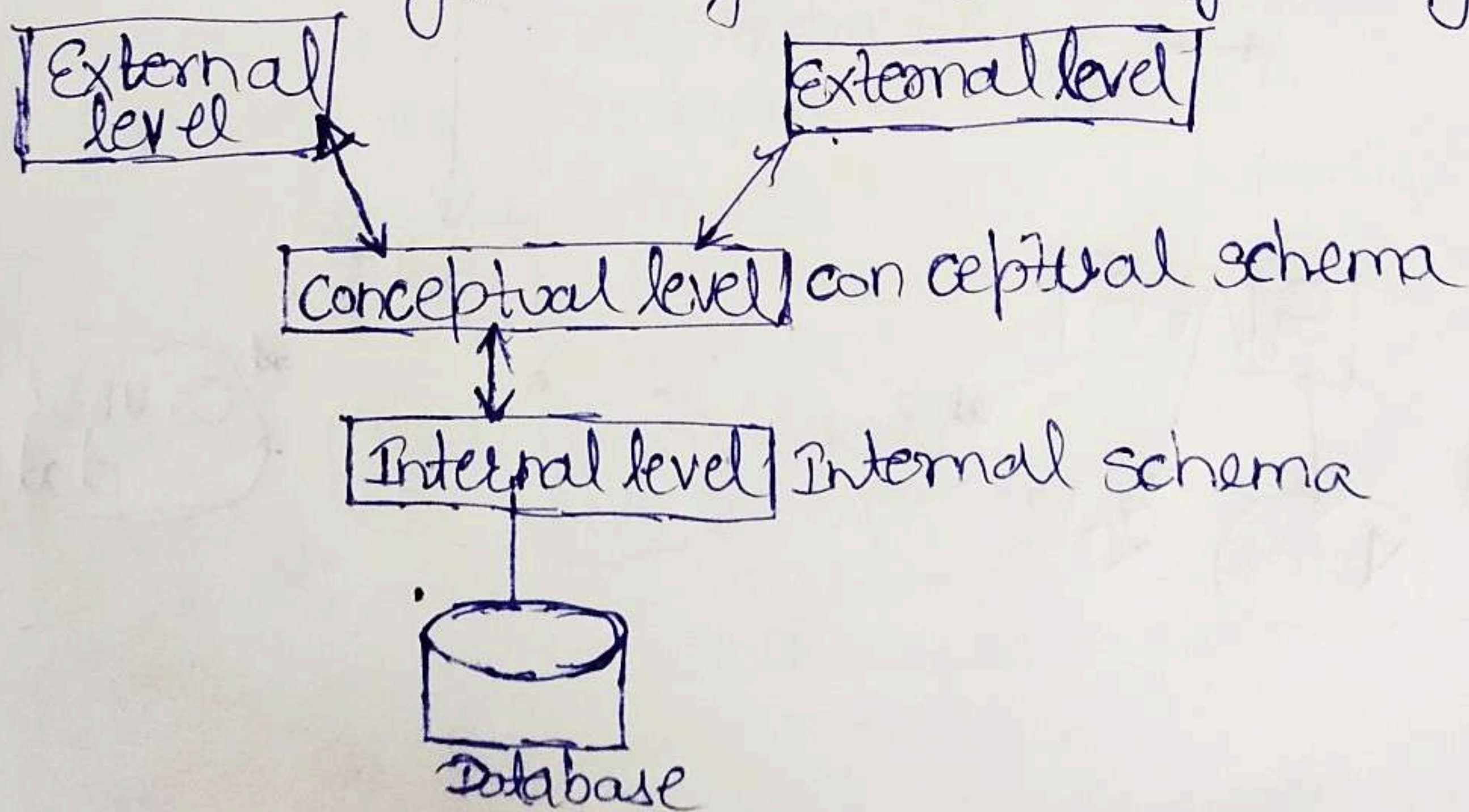
- describes how individual users view data.
- Includes tables, attributes
- different users can have diffn views.
- Hides the rest of the database.

(ii) Conceptual Schema (logical level):

- describes the overall structure of the database
- Include tables, attributes, relationship, constraints
- Does not include physical storage details.

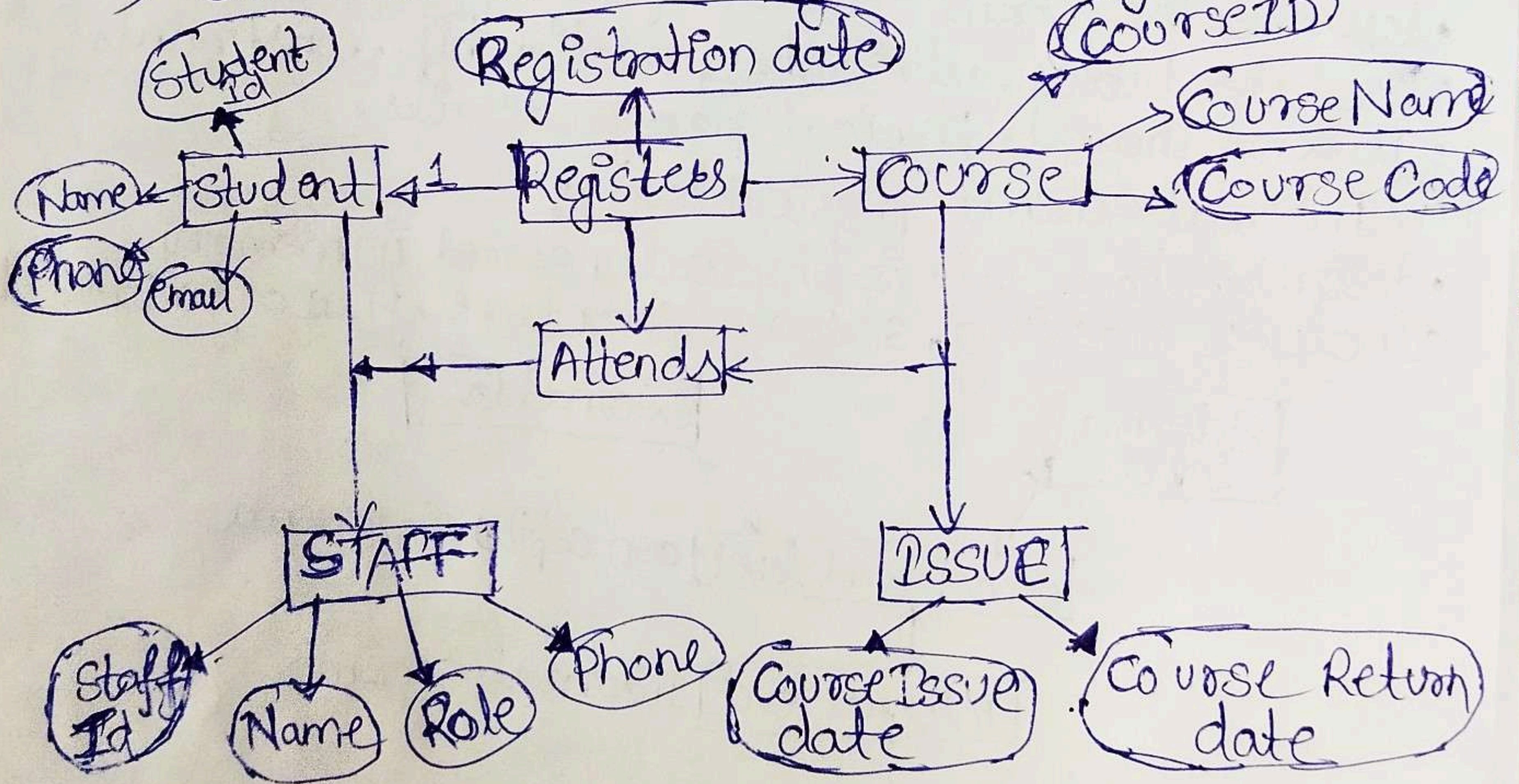
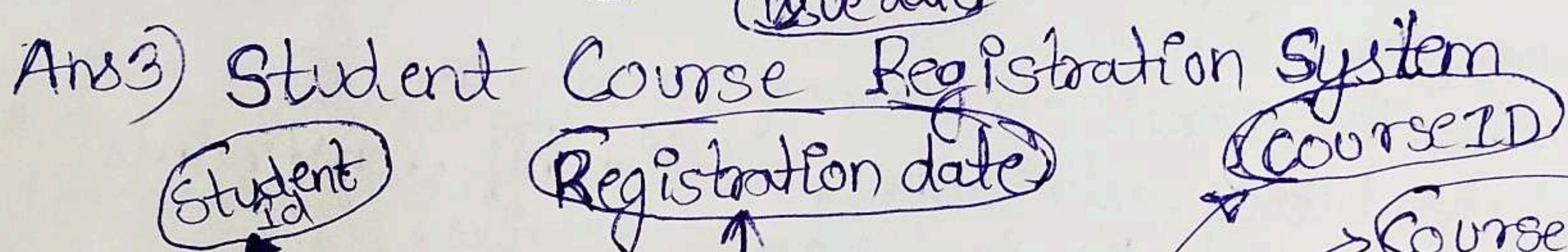
(iii) Internal Schema (physical level):

- describes how data is physically stored in memory.
- Includes indexing, storage structure, file organization etc.




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graph TD
    Librarian[Librarian] --- ms2_login[ms2) login]
    Librarian --- Password[Password]
    Librarian --- Library[Library]
  
```



Section - C

1) In this, Entities are Patient and its Attributes are Patient ID, Name, Age, Gender, Address, Phone and A doctor is entity and its attribute are doctor ID, Name, Speciality, Phone and Email and one more entity Appointment and its attribute are Appointment ID, Appointment date, Appointment time and here foreign keys are patient ID which link to patient and Doctor ID which links to doctor. And here relationship are between patients and appointment because A patient can have many appointments and its cardinality is 1:M and another relationship is between doctor and appointment. ~~A doctor can have many appointments and its cardi~~

Q2) Entities key

1. Customer - Customer ID (PK), Name, Email (unique), Phone (unique), Address

2. Product - Product ID (PK), Name, Price

3. Order - order ID (PK), order Date, Customer ID (FK)

4. Order Item - order Item ID (PK), order ID (FK → order)

Product ID (FK), Quantity, Price.

Relationship are between customer and order which is 1:M and order and Product which is M:M

Here Constraints are Primary key must be unique for each entity and Foreign key maintains relationship and email and Phone must be unique and check price should be greater than zero.

Q3) Data abstraction helps by ^{hiding} unnecessary details from user and showing only essential parts to user

1) physical level (lowest level) :- shows how data is stored physically on disk, like files, indexes.

eg :- The database stores student records block on a server

2) logical level (conceptual level) - shows what data consists and how it is related without worrying about physical storage.

ex:- Tables student (ID, Name, Age, department) and course (courseId, Name) and their relationship.

~~view~~ view level (External level): shows only the part of data a user needs. Diffn users see different views.

ex: A professor see student grades in their courses and A student see only their own grades and courses.

1) In this my entities ^{section - D} are customer, order and product